

Brock Biology of Microorganisms, 15e (Madigan et al.)
Chapter 15 Functional Diversity of Microorganisms

15.1 Multiple Choice Questions

1) Microbial diversity is thought of in terms of phylogenetic diversity and functional diversity because

- A) similar functional traits are often found in divergent phylogenetic groups.
- B) phylogenetic diversity dictates functional diversity in microbes.
- C) functional diversity only pertains to functions that are not phylogenetic.
- D) similar phylogenetic groups almost always have similar functional traits.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.1

2) Functional traits can be shared between organisms with divergent SSU rRNA gene sequences because

- A) functional traits may be lost by different lineages over time.
- B) functional traits may evolve independently in different lineages over time.
- C) horizontal gene transfer can result in the transfer of genes between divergent lineages.
- D) functional traits may evolve independently, be shared through horizontal gene transfer, or be lost in divergent lineages.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.1

3) What form of photosynthesis evolved first on Earth?

- A) oxygenic
- B) anoxygenic
- C) autotrophic
- D) heterotrophic

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.2

4) The only group of oxygenic phototrophic bacteria are the

- A) green nonsulfur bacteria.
- B) *Chloroflexi*.
- C) *Cyanobacteria*.
- D) *Chlorobi*.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.3

5) All microbial phototrophs

A) fix carbon dioxide.

B) use water as an electron donor for photosynthesis.

C) fix carbon dioxide and use water as an electron donor for photosynthesis.

D) use light energy to drive electron transport to generate energy.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.2

6) Which of the following conditions would you use to enrich purple nonsulfur bacteria from lake sediments?

A) anoxic media with a good light source

B) mineral media with ammonia and oxygen in the dark

C) anoxic mineral media with an organic acid and a good light source

D) mineral media with nitrate and acetate in the dark

Answer: C

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 15.5

7) Aerobic anoxygenic phototrophs

A) do not use bacteriochlorophyll.

B) are an ancient monophyletic group of bacteria.

C) can be either autotrophic or heterotrophic.

D) cannot fix CO₂.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.5

8) Aside from using light as a source of energy, another important part of the functional diversity of some bacterial phototrophs is their ability to

A) bioluminesce.

B) form prosthecate and stalked cells.

C) fix nitrogen gas.

D) degrade neurotoxins.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.12

9) Which group of bacteria would you expect to find in a shallow clear hot spring with high concentrations of geochemically produced hydrogen sulfide?

A) purple sulfur bacteria

B) single-celled *Cyanobacteria*

C) dissimulative sulfate-reducing bacteria

D) symbiotic diazotrophs

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 15.4

10) The ecological diversity of phototrophic bacteria is illustrated by the different _____ that allow them to perform photosynthesis in low light environments such as deep in the ocean, in various sediments, and in thick layered mats.

- A) nitrogen fixation strategies
- B) bacteriochlorophyll and photosystems
- C) specialized stalked cells
- D) endospores and resistant cell types

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 15.6

11) You are looking at a sample from the top green layer of a colorful bacterial mat. Using phase contrast and fluorescence microscopy, you see evenly spaced cells in the middle of a filament that are slightly smaller than the others. The smaller cells are not fluorescent, indicating they lack photosystem II, but the rest of the filament is fluorescent. What type of filamentous bacteria are you most likely looking at?

- A) purple nonsulfur bacteria
- B) green sulfur bacteria
- C) *Acidobacteria*
- D) *Cyanobacteria*

Answer: D

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 15.3

12) You are looking at a sample from the top green layer of a colorful bacterial mat. Using phase contrast and fluorescence microscopy, you see evenly spaced cells in the middle of a filament that are slightly smaller than the others. The smaller cells are not fluorescent, indicating they lack photosystem II, but the rest of the filament is fluorescent. What is the most likely function of the smaller cells in the filament?

- A) survival in the dark
- B) nitrogen fixation
- C) sulfur storage
- D) nitrate storage

Answer: B

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 15.3

13) Sulfate-reducing bacteria are a _____ that are found in _____ environments.

- A) monophyletic group / anoxic
- B) polyphyletic group / oxic
- C) morphologically distinct group / anoxic
- D) morphologically distinct group / oxic

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.10

14) Typical electron donors used by dissimilative sulfate-reducers are

- A) hydrogen, water, and hydrogen sulfide.
- B) hydrogen, lactate, and pyruvate.
- C) carbon dioxide, hydrogen, and water.
- D) sulfate, sulfur, and thiosulfate.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.10

15) Considering the relationship between oxidizers and reducers, which of the following pairs of microorganisms are NOT likely to form close associations or consortia?

- A) purple sulfur bacteria and dissimilative sulfur-reducers
- B) dissimilative sulfur-reducers and dissimilative sulfur-oxidizers
- C) diazotrophs and fungi
- D) nitrogen-fixing cyanobacteria and dissimilative sulfate-reducers

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 15.6

16) The vacuole of the giant sulfide-oxidizing bacterium *Thiomargarita namibiensis* contains _____ that is used to oxidize hydrogen sulfide under _____ conditions.

- A) nitrate / anoxic
- B) sulfate / anoxic
- C) oxygen / oxic
- D) sulfur / anoxic

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.11

17) Nitrogen fixation is

- A) rare among microorganisms.
- B) widespread among microorganisms.
- C) limited to *Proteobacteria*.
- D) common in *Bacteria*, but absent in *Archaea*.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.12

18) The phylogenetic trees of the *nifH* and SSU rRNA genes of nitrogen-fixing bacteria are incongruent because

- A) nitrogenase is an anaerobic enzyme.
- B) this is a characteristic of all enzyme genes.
- C) of horizontal gene transfer.
- D) very few microbes have the *nifH* gene.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.12

19) The *Cyanobacteria* are united by their _____, but their _____ is very diverse.

- A) physiology / morphology
- B) morphology / physiology
- C) ecology / physiology
- D) SSU rRNA gene sequences / physiology

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.3

20) Purple sulfur bacteria are

- A) alphaproteobacteria that oxidize hydrogen sulfide in the presence of oxygen or nitrate.
- B) gammaproteobacteria that use hydrogen sulfide as an electron donor for photosynthesis.
- C) a phylogenetically heterogeneous group of microbes that reduce sulfate to sulfur.
- D) thermophilic bacteria that reduce elemental sulfur to sulfide.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.4

21) Green sulfur bacteria are often found at the greatest depths of all phototrophic cells in lakes, oceans, and microbial mats. This is because green sulfur bacteria

- A) are very sensitive to UV irradiation, despite their reliance on light for energy.
- B) are the only phototrophs that can use hydrogen sulfide as an electron donor for photosynthesis.
- C) are strict anaerobes.
- D) have chlorosomes that are very efficient at harvesting light, allowing them to perform photosynthesis where light intensities are very low.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 15.6

22) Dissimilatory sulfide-oxidizers use a variety of unique ecological strategies in order to

- A) avoid the toxicity of hydrogen sulfide.
- B) position themselves at interfaces where sulfide and oxygen meet.
- C) position themselves where there is sufficient light for sulfide oxidation.
- D) find organic carbon substrates.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.11

23) *Thioploca* species contain large vacuoles that store _____, which is used to _____ under anaerobic conditions.

- A) nitrate / oxidize sulfide
- B) oxygen / oxidize sulfur
- C) sulfur / reduce nitrate
- D) glycogen / provide energy

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.11

24) In symbiotic associations between sulfur chemolithotrophs and eukaryotes such as tube worms and giant clams, the bacterial symbiont provides _____ while the host provides _____.

- A) nitrogen / carbon and oxygen
- B) sulfide / oxygen
- C) a carbon and energy source / stable sulfide and oxygen levels
- D) organic acids / stable sulfate levels

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.11

25) The *Bacteria* and *Archaea* are the ONLY domains in which organisms exist that can

- A) catalyze the transformation of inorganic nitrogen species for biosynthesis.
- B) assimilate nitrite.
- C) conserve energy from the transformation of inorganic nitrogen species.
- D) catalyze the incorporation of inorganic nitrogen species into biomolecules.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.12

26) Nitrogenase is a critical enzyme in the nitrogen cycle because it catalyzes the

- A) assimilation of ammonia.
- B) dissimilative reduction of nitrate to nitrogen gas.
- C) fixation of nitrogen gas into ammonia.
- D) oxidation of nitrogen gas into nitrate.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.12

- 27) Diazotrophs maximize nitrogenase activity by
- A) closely coupling N₂ fixation with high rates of photosynthesis.
 - B) fixing N₂ at times when oxygen is absent or in low concentrations.
 - C) producing specialized structures to protect nitrogenase from oxygen.
 - D) producing specialized structures or fixing N₂ at times when oxygen is absent to protect nitrogenase from oxygen.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.12

- 28) What growth conditions would you use to enrich for ammonia-oxidizing bacteria?

- A) aerobic mineral media with ammonia and bicarbonate incubated in the dark
- B) anaerobic mineral media with acetate and ammonia incubated in the dark
- C) aerobic mineral media with acetate and ammonia incubated in the dark
- D) anaerobic mineral media with ammonia and nitrate incubated in the dark

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 15.13

- 29) *Escherichia coli* is a chemoorganotrophic, facultative aerobe and belongs to the phylum *Proteobacteria*. Based on this information, which of the following functional groups does *E. coli* most likely belong to?

- A) green sulfur bacteria
- B) ammonia-oxidizing bacteria
- C) denitrifying bacteria
- D) nitrite-oxidizing bacteria

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 15.13

- 30) Which of the following traits do ammonia and nitrite oxidizers share?

- A) chemoorganotrophic
- B) autotrophic
- C) aerobic
- D) autotrophic and aerobic

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.13

- 31) Dissimilative iron-reducing bacteria couple which oxidation-reduction reactions?

- A) The reduction of ferric iron or manganese and the oxidation of sulfur.
- B) The reduction of ferric iron or manganese and the fermentation of sugars.
- C) The oxidation of H₂ or organic compounds and the reduction of ferric iron or manganese.
- D) The reduction of ferric iron and photosynthesis.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.14

32) In which of the following environments would you expect to find a high number of iron-reducing bacteria?

- A) acidic mine drainage
- B) deep subsurface
- C) open ocean
- D) agriculture soil

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 15.14

33) Which of the following functional groups would you expect to find in aerobic iron- and sulfur-rich acidic mine drainage?

- A) dissimilative iron-oxidizers
- B) dissimilative iron-reducers
- C) purple sulfur bacteria
- D) purple sulfur bacteria and dissimilative iron-reducers

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 15.15

34) What is one especially unique characteristic of methanotrophs?

- A) They metabolize H₂S.
- B) They have unusually large amounts of sterols.
- C) They can exist as consortia.
- D) They are capable of bioluminescence.

Answer: B

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 15.16

35) What microbes have a symbiotic association with mussels, living in their gill tissue?

- A) methanotrophs
- B) iron-oxidizing bacteria
- C) bioluminescent bacteria
- D) sulfur oxidizing bacteria

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.16

36) What is the difference between a methylotroph and methanotroph?

- A) Methylotrophs are strict anaerobes, while methanotrophs are strict aerobes.
- B) Methylotrophs use organic carbon lacking C-C bonds as electron donors and carbon sources, while methanotrophs exclusively use methane as an electron donor and carbon source.
- C) Methylotrophs produce organic carbon lacking C-C bonds, while methanotrophs exclusively produce methane.
- D) Methylotrophs are in *Proteobacteria*, while methanotrophs are in *Archaea*.

Answer: B

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 15.16

37) Most methanotrophs are obligate _____ because _____.

- A) aerobes / the initial step of methane oxidation requires O₂
- B) anaerobes / methane is produced in anaerobic environments
- C) fermenters / methane only provides enough energy for fermentation
- D) heterotrophs / methane does not provide enough carbon for growth

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.16

38) Which of the following functional groups form symbiotic relationships with Eukaryotes in deep sea environments?

- A) dissimilatory iron-oxidizers
- B) methanotrophs
- C) sulfide-oxidizers
- D) methanotrophs and sulfide-oxidizers

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.16

39) Epibiotic predators

- A) obtain nutrients by attaching to the surface of prey.
- B) invade the cytoplasm of their prey.
- C) invade the periplasmic space of their prey.
- D) find prey using swarming behavior.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.17

40) *Bdellovibrio* species are _____ cells that replicate in the _____.

- A) symbiotic / gills of clams
- B) lithotrophic / deep subsurface
- C) predatory / periplasmic space
- D) denitrifying / sediments of freshwater lakes

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.17

41) *Myxobacteria* display _____ behavior and form _____ that produce myxospores.

- A) gliding / slime trails
- B) chemotactic / colonies
- C) phototactic / slime trails
- D) swarming / fruiting bodies

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.17

42) Bioluminescence develops when

- A) population density is high.
- B) oxygen is depleted.
- C) ambient light is low.
- D) the host induces the bioluminescence genes.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.18

43) Bioluminescence is catalyzed by luciferase, which creates light by

- A) shunting electrons from FMNH₂ to O₂, which releases energy in the form of light.
- B) reversing photosynthesis, which produces a surplus of ATP.
- C) hydrolyzing ATP to release the excess energy as light.
- D) reducing LuxR, the regulatory protein that controls the lux operon.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.18

44) Bioluminescence is controlled by the gene regulatory mechanism called quorum sensing, wherein the

- A) depletion of oxygen is detected by an inducer molecule which activates transcription of the *lux* operon (genes *luxCDABE*).
- B) accumulation of an inducer molecule activates transcription of the *lux* operon (genes *luxCDABE*).
- C) absence of light activates an inducer molecule which activates transcription of the *lux* operon (genes *luxCDABE*).
- D) host tissues produce an autoinducer molecule which activates transcription of the *lux* operon (genes *luxCDABE*).

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.18

45) Bioluminescent microorganisms are found in which of the following groups?

- A) *Firmicutes*
- B) *Thaumarchaeota*
- C) *Gammaproteobacteria*
- D) *Verrucomicrobia*

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.18

46) Prosthecae are an adaptation for

- A) anaerobic growth.
- B) cell differentiation under harsh conditions.
- C) nitrogen fixation.
- D) low nutrient aquatic environments.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.20

15.2 True/False Questions

1) Although *Cyanobacteria* contribute more than 35% of all photosynthesis on Earth, they are not visible in the natural environment.

Answer: FALSE

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 15.3

2) Akinetes and hormogonia are specialized cells, found in some cyanobacteria, that facilitate nitrogen fixation.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.3

3) Unicellular cyanobacteria are the most abundant phototrophs in the ocean and perform 80% of marine photosynthesis.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.3

4) Diazotrophs are found in all three domains of life.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.12

5) Diazotrophs are obligately symbiotic bacteria that fix N_2 into ammonia, thus providing a critical source of nitrogen to their host.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.12

6) There are currently no known predatory bacteria that are adapted to invade or kill nearby cells.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.17

7) Nitrogenase is an oxygen-sensitive enzyme even though many diazotrophs are aerobic microorganisms.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.12

8) Denitrifying bacteria are often facultative anaerobes, capable of performing other forms of respiration or fermentation as well as nitrate reduction.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.13

9) Dissimilatory nitrate reduction results in the production of gaseous products, while assimilatory nitrate reduction results in the production of ammonia for biosynthesis.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.13

10) Iron-reducing bacteria are obligately aerobic organisms that reduce ferric iron and oxidize hydrogen or organic compounds.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.14

11) The ecological diversity of microbes is largely dependent on their complex morphological and structural diversity.

Answer: FALSE

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 15.1

12) Dissimilatory iron-oxidation is a relatively new metabolism and is restricted to one unusual phylum of bacteria.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.15

13) Sulfide oxidizers are among the most ecologically diverse trophic groups because of the different strategies they employ to access sufficient amounts of sulfide and oxygen simultaneously.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.11

14) Anoxygenic phototrophs may or may not be autotrophs.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.8

15) All phototrophic bacteria use a monophyletic photosystem in photosynthesis, thus there is intense competition for light between different phototrophic bacteria.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.7

16) Organisms that fix dinitrogen gas into ammonia are called diazotrophs.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.12

17) Spirilla and spirochetes are closely related bacteria that exhibit a unique corkscrew motility.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.19

18) Budding bacteria form new cell wall material at a single point rather than throughout the whole cell.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.20

19) Bacteria that contain complex internal membrane systems are more likely to divide by budding than by binary fission.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.20

20) Prosthecae are thought to be an adaptation to growth in nutrient-rich environments such as sediments and sewage.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.20

21) *Bacteria* and *Archaea* are general single-celled organisms, but some phyla form sheaths around multiple cells to facilitate attachment and filament formation.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.21

22) Magnetosomes are specialized magnetic structures, found in iron-reducing bacteria, that aid in chemotaxis towards iron-oxides.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.22

23) Bioluminescence is a social behavior of bacterial cells that causes luminescence to occur when a dense cell population has been attained.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.18

24) Budding is a form of reproduction whereby prokaryotic microbes can reproduce sexually.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 15.20

15.3 Essay Questions

1) Discuss the ecological importance of the enzyme nitrogenase, the diversity and physiology of the various groups of organisms that contain it, and the strategies these organisms use to optimize conditions for the enzyme.

Answer: The enzyme nitrogenase is the only enzyme that can catalyze the conversion of nitrogen gas into ammonia, thus it is critical to the N cycle in all ecosystems. *Bacteria* and *Archaea* from many different phyla contain *nifH* (the gene that encodes for the nitrogenase enzyme). Some of the microbes form symbiotic associations with plants and fungi, while others are free-living. The ability to fix nitrogen is not restricted to one particular metabolic group of microbes and nitrogenase can be found in aerobic, anaerobic, heterotrophic, autotrophic, and phototrophic microbes. Nitrogenase is inactivated by oxygen. Microbes that live in aerobic environments or produce oxygen themselves use a variety of strategies to keep nitrogenase and oxygen separate. Examples include specialized cells (heterocysts) that are non-photosynthetic and impermeable to oxygen, production of a thick polysaccharide capsule to prevent oxygen diffusion into the cell, and induction of nitrogenase at night or at times when oxygen concentrations are low.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 15.12

2) Compare and contrast two different ecological strategies that sulfide-oxidizing bacteria use to access sulfide and oxygen.

Answer: Answer could explain any of the strategies outlined in Section 14.11 (on pages 451 and 452) in the text. In general sulfide oxidizers use specialized attachment, motility, or storage strategies to access sulfide and oxygen (or nitrate).

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 15.11

3) Tilling or plowing is a common agriculture practice that has been proven to have a profound impact on the nitrogen cycle. Tilling breaks up hard soils to ease the planting and germination of crops, and it also introduces more oxygen into the soil. Many fertilizers contain N in the form of ammonia because ammonia is readily absorbed and used by plants, while N in the form of nitrate is more mobile and is more easily lost from the soil. Based on your knowledge of microbial transformations of inorganic nitrogen species, what impact would you predict extensive tilling to have on the fate of ammonium-based nitrogen fertilizer?

Answer: Ammonia-oxidizing and nitrite-oxidizing bacteria are found in soils and together convert ammonia to nitrate. Nitrate can be lost from the soil via leaching or denitrification. Losing nitrate from the soil represents a loss of total N from the soil, decreasing soil fertility. Introduction of more oxygen into soil through tilling would most likely increase the rate of the nitrification, resulting in increased N loss from the soil. Reduced tilling may help retain fertilizer and natural N in a soil and increase soil fertility.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 15.13

4) *Thiobacillus* species are acidophilic chemolithotrophs capable of autotrophic growth. They perform the dissimilative oxidation of H_2S , S^0 , thiosulfate, or Fe^{+2} . Diagram or explain how *Thiobacillus* species obtain carbon and energy for growth and predict what type of environment they would be found in.

Answer: *Thiobacillus* species obtain carbon for growth from carbon dioxide, thus they do not require organic carbon-such as sugars-for growth. Energy is generated by the transfer of electrons from H_2S , S^0 , thiosulfate, or Fe^{+2} to oxygen. The transfer of electrons through an electron transport chain generates a proton motive force that subsequently drives ATP production. The metabolism makes them well-adapted for acidic, aerobic environments that are high in reduced sulfur and iron species, such as acid mine drainage or seeps and springs that contain the reduced chemical species that they use as electron donors. The environments they survive in do not need to have a source of organic carbon, but they do need to be aerobic.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 15.11

5) Based on the FUNCTIONAL characteristics of the *Cyanobacteria* and *Proteobacteria* phyla, which phylum do you think is more deeply rooted in the tree of life (i.e., which phylum branches off closer to the last universal common ancestor)? Use at least two pieces of evidence to support your answer. How could you test your hypothesis?

Answer: *Proteobacteria* are more likely to be deeply rooted in the tree of life compared to *Cyanobacteria* for two reasons. First, *Proteobacteria* are very functionally diverse and include anoxygenic phototrophs, sulfur-oxidizers, sulfur-reducers, and a huge array of other anaerobic and aerobic physiologies. In comparison, *Cyanobacteria* are exclusively oxygenic phototrophs which could indicate that this phylum is younger and has had less time to evolve diverse physiologies. Second, conditions on early Earth were anaerobic and highly reducing. The physiology of many *Proteobacteria* indicates that ancestors of this group evolved during this time and have retained the ability to utilize a diverse array of reduced compounds for energy. In addition, many anoxygenic phototrophs are *Proteobacteria*. Anoxygenic photosynthesis evolved prior to oxygenic photosynthesis, which also supports the idea that *Proteobacteria* may be more deeply rooted than *Cyanobacteria*. To test this hypothesis, comprehensive phylogenetic trees including many bacterial phyla could be constructed based on SSU rRNA sequences and sequences of other housekeeping genes to determine which phylum is more deeply rooted based on molecular sequences that are not involved in energy generation.

Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 15.2

6) How do *Cyanobacteria* impact the ecology of both aquatic and terrestrial environments on Earth?

Answer: *Cyanobacteria* can contribute up to 80% of photosynthesis in some aquatic environments. They are also especially important in extreme terrestrial environments that are characterized by high salinity, extremely low moisture, extreme temperatures, and/or high UV irradiation. For example, they are the main photosynthetic organisms in photosynthetic soil crusts in the deserts and in the dry valleys of Antarctica. In addition to contributing to primary productivity, many species are also able to fix nitrogen. The addition of both nitrogen and carbon to ecosystems is critical for productivity. Their ability to fix nitrogen (which eukaryotic organisms CANNOT do on their own) increases primary productivity in environments with very low N.

Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 15.3

7) Design an enrichment strategy for the isolation of a heterotrophic sulfate-reducer that could help clean up the oil spill in the sediments of the Gulf of Mexico.

Answer: To enrich for heterotrophic sulfate-reducers, the media must contain an organic source of carbon, a source of N, P, and trace elements, and sulfate. The growth conditions must also be anaerobic, as most sulfate-reducers are obligate anaerobes and marine sediments are mostly depleted or devoid of oxygen. The carbon source should be hydrocarbons similar to those found in crude oil or very small amounts of crude oil. Lastly, the bottle or tubes should be incubated in the dark to prevent the growth of anoxygenic phototrophs.

Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 15.9

8) What functional group (or groups) of microbes would be most useful in a microbial fuel cell in which electrons are transferred to and from solid metal electrodes? Why?

Answer: Iron-reducers and iron-oxidizers would be most useful in this type of microbial fuel cell, as they are adapted to transfer electrons to and from solid electron acceptors and donors. Most terminal oxidases and other electron transport proteins are inside the cell wall in the cytoplasmic membrane where they cannot contact solid surfaces. Iron-reducers in particular have electron transport proteins embedded in the outer membrane where they can effectively transfer iron to and from solid redox active surfaces (such as metal electrodes).

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 15.15

9) What is convergent evolution and how is it different from horizontal gene transfer?

Answer: Convergent evolution is the evolution of the same or similar function that occurred independently more than once. Organisms that have evolved the same function through convergent evolution utilize different enzymes or structures to perform the same function. Horizontal gene transfer is the transfer of particular genes or set of genes that confer the same function on another group of microorganisms. Both processes result in divergent species (species lacking a recent common ancestor) performing the same function.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 15.1

10) Given that multiple diverse phyla (e.g., diverse phylogenetic groups) of microorganisms can perform similar functions in ecosystems, how we might predict the microbial functions performed by a particular organism or in a particular environment without relying on growing all of the microbes in the lab?

Answer: Without growing all of the microbes in the environment, we could rely on molecular (nucleic acid based) techniques to predict the function of uncultivated microbes. Identifying microbes based on their SSU rRNA sequence is not always predictive of function because of convergent evolution, horizontal gene transfer, and gene loss; therefore, techniques that look for and quantify genes known to be involved in particular functions are more reliable and informative. This is one reason that metagenomics and metatranscriptomics are now being used to study microbial communities (Chapter 10).

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 15.1

11) How are anoxygenic phototrophic microorganisms more physiologically diverse than oxygenic phototrophic microorganisms? Be specific in your answers and use examples to illustrate your conclusions.

Answer: Oxygenic phototrophs (*Cyanobacteria*) are only capable of using water as an electron donor for photosynthesis. Anoxygenic phototrophs can often use multiple electron donors, such as sulfide, sulfur, ferrous iron, and organic compounds. In addition, all *Cyanobacteria* are autotrophs, while some anoxygenic phototrophs are capable of both autotrophic and heterotrophic growth. The ability to grow heterotrophically enables many anoxygenic phototrophs to grow without light. Anoxygenic phototrophs also have a variety of different photosystems and pigments that help them harvest different wavelengths of light at greater depths than oxygenic phototrophs.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 15.8